

Voyager 1

Voyager 1 is a space probe launched by NASA on September 5, 1977, as part of the Voyager program, to study the outer Solar System and the interstellar space beyond the Sun's heliosphere. It was launched 16 days after its twin, *Voyager 2*. It communicates through the NASA Deep Space Network (DSN) to receive routine commands and to transmit data to Earth. Real-time distance and velocity data are provided by NASA and JPL.^[4] At a distance of 172.59 AU (25.8 billion km; 16.0 billion mi) as of March 2026,^[4] it is the most distant human-made object from Earth.^[5] *Voyager 1* is also projected to reach a distance of one light day from Earth in November 2026.^[6]

The probe made flybys of Jupiter, Saturn, and Saturn's largest moon, Titan. NASA had a choice of either conducting a Pluto or Titan flyby. Exploration of Titan took priority because it was known to have a substantial atmosphere.^{[7][8][9]} *Voyager 1* studied the weather, magnetic fields, and rings of the two gas giants and was the first probe to provide detailed images of their moons.

As part of the Voyager program and like its sister craft *Voyager 2*, the spacecraft's extended mission is to locate and study the regions and boundaries of the outer heliosphere and to begin exploring the interstellar medium. *Voyager 1* crossed the heliopause and entered interstellar space on August 25, 2012, making it the first spacecraft to do so.^{[10][11]} Two years later, *Voyager 1* began experiencing a third wave of coronal mass ejections from the Sun that continued to at least December 15, 2014, further confirming that the probe is in interstellar space.^[12]

In 2017, the Voyager team successfully fired the spacecraft's trajectory correction maneuver (TCM) thrusters for the first time since 1980, enabling the mission to be extended by two to three years.^[13] *Voyager 1* experienced successful revivals of several thrusters in 2018, 2019, and 2025.^[14]

Voyager 1's extended mission is expected to continue to return scientific data for several more years.^[15] Its radioisotope thermoelectric generators (RTGs) may supply enough electric power to return engineering data until 2036.^[16] As of 2026, only two instruments are operational, the Plasma Wave Subsystem and magnetometer.

Mission background

A 1960s proposal for a Grand Tour to study the outer planets led NASA to begin work on a mission during the early 1970s.^[17] Initially, *Voyager 1* was planned as *Mariner 11* of the Mariner program. Due to budget cuts, the mission was reduced to a flyby of Jupiter and Saturn and renamed the Mariner Jupiter-Saturn probes. The name was changed to *Voyager* when the probe designs began to differ substantially from Mariner missions.^[18]

	Voyager 1
 Artist's rendering of the <i>Voyager</i> spacecraft	
Mission type	Outer planetary, heliosphere, and interstellar medium exploration
Operator	NASA / Jet Propulsion Laboratory
COSPAR ID	1977-084A ^[1]
SATCAT no.	10321 ^[1]
Website	science.nasa.gov/mission/voyager/ (https://science.nasa.gov/mission/voyager/)
Mission duration	Total: 48 years, 263 days (elapsed) Planetary mission: 3 years, 3 months, 9 days Interstellar mission: 45 years, 163 days (elapsed)
Spacecraft properties	
Spacecraft type	Mariner Jupiter-Saturn
Manufacturer	Jet Propulsion Laboratory
Launch mass	815 kg (1,797 lb) ^[2]
Dry mass	721.9 kg (1,592 lb) ^[3]
Power	470 watts (at launch)
Start of mission	
Launch date	September 5, 1977, 12:56:01 UTC
Rocket	Titan IIIE
Launch site	Cape Canaveral Launch Complex 41
End of mission	
Flyby of Jupiter	
Closest approach	March 5, 1979
Distance	349,000 km (217,000 mi)
Flyby of Saturn	

Spacecraft components

Voyager 1 was built by the Jet Propulsion Laboratory (JPL). It has a bus shaped like a decagonal (ten-sided) prism. It has 16 hydrazine thrusters, three-axis stabilization gyroscopes, and referencing instruments to keep the probe's radio antenna pointed toward Earth. Collectively, these instruments are part of the Attitude and Articulation Control Subsystem (AACS), along with redundant units of most instruments and eight backup thrusters.^[19] The spacecraft also included 11 scientific instruments to study celestial objects such as planets as it travels through space.^[20]

Communication system

The radio communication system of *Voyager 1* was designed to be used up to and beyond the limits of the Solar System. It has a 3.7-meter (12 ft) diameter high-gain Cassegrain antenna to send and receive radio waves via the three Deep Space Network stations on the Earth.^[21] The spacecraft normally transmits data to Earth over Deep Space Network Channel 18, using a frequency of either 2.3 GHz or 8.4 GHz, while signals from Earth to Voyager are transmitted at 2.1 GHz.^[22]

When *Voyager 1* is unable to communicate with the Earth, its digital tape recorder (DTR) can record about 64 megabytes of data for later transmission.^[23] As of 2025, signals from *Voyager 1* took more than 23 hours to reach Earth.^[4]

Power

Voyager 1 has three radioisotope thermoelectric generators (RTGs) mounted on a boom. Each MHW-RTG contains 24 pressed plutonium-238 PuO₂ oxide spheres.^[24] The RTGs generated about 470 W of electric power at the time of launch, with the remainder being dissipated as waste heat.^[25] The power output of the RTGs declines over time due to the 87.7-year half-life of the fuel and degradation of the thermocouples. According to current plans, they may continue operating with at least one science instrument into the 2030s.^{[20][24]}

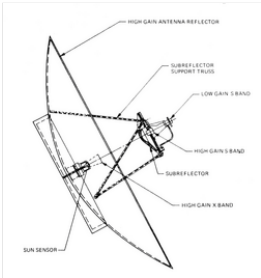
Computers

Unlike *Voyager's* other instruments, the operation of the cameras for visible light is not autonomous, but is controlled by an imaging parameter table contained in one of the digital computers, the Flight Data Subsystem (FDS). Since the 1990s, most space probes have been equipped with completely autonomous cameras.^[26]

The computer command subsystem (CCS) controls the cameras. The CCS contains fixed computer programs, such as command decoding, fault-detection and fault-correction routines, antenna pointing routines, and spacecraft sequencing routines. This computer is an improved version of the one that was used in the 1970s Viking orbiters.^[27]

The Attitude and Articulation Control Subsystem (AACS) controls the spacecraft orientation. It keeps the high-gain antenna pointing towards Earth, controls attitude changes, and points the scan platform. The custom-built AACS systems on both Voyagers are the same.^{[28][29]}

Closest approach	November 12, 1980
Distance	124,000 km (77,000 mi)
Flyby of Titan (atmosphere study)	
Closest approach	November 12, 1980
Distance	6,490 km (4,030 mi)
Instruments	
Large Strategic Science Missions	
Planetary Science Division	
Voyager program	



The 3.7 m (12 ft) diameter high gain dish antenna used on the Voyager craft

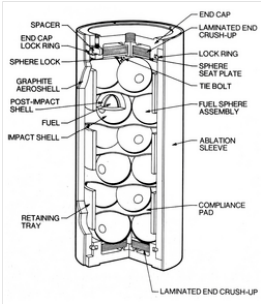
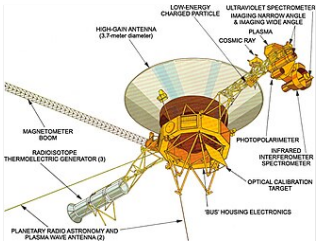


Diagram of RTG fuel container, showing plutonium-238 oxide spheres

Scientific instruments



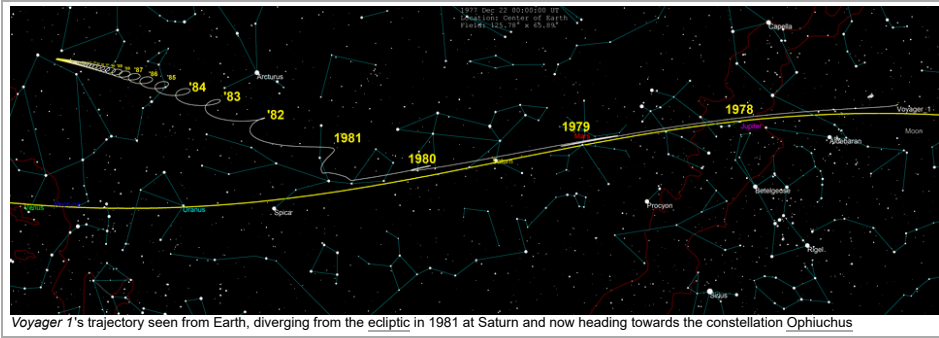
Locations of Voyager's scientific instruments

Instrument name	Abbr.	Description																																																																																						
Imaging Science System (disabled)	(ISS)	Used a two-camera system (narrow-angle/wide-angle) to provide images of Jupiter, Saturn and other objects along the trajectory. Off to save power (February 14, 1990)																																																																																						
		<table><tr><th colspan="8">Filters</th></tr><tr><th colspan="4">Narrow-angle camera^[30]</th><th colspan="4">Wide-angle camera^[31]</th></tr><tr><th>Name</th><th>Wavelength</th><th>Spectrum</th><th>Sensitivity</th><th>Name</th><th>Wavelength</th><th>Spectrum</th><th>Sensitivity</th></tr><tr><td>0 – Clear</td><td>280–640 nm</td><td></td><td></td><td>2 – Clear</td><td>280–640 nm</td><td></td><td></td></tr><tr><td>4 – Clear</td><td>280–640 nm</td><td></td><td></td><td>3 – Violet</td><td>350–450 nm</td><td></td><td></td></tr><tr><td>7 – UV</td><td>280–370 nm</td><td></td><td></td><td>1 – Blue</td><td>430–530 nm</td><td></td><td></td></tr><tr><td>1 – Violet</td><td>350–450 nm</td><td></td><td></td><td>6 – CH₄-U</td><td>536–546 nm</td><td></td><td></td></tr><tr><td>2 – Blue</td><td>430–530 nm</td><td></td><td></td><td>5 – Green</td><td>530–640 nm</td><td></td><td></td></tr><tr><td>5 – Green</td><td>530–640 nm</td><td></td><td></td><td>4 – Na-D</td><td>588–590 nm</td><td></td><td></td></tr><tr><td>6 – Green</td><td>530–640 nm</td><td></td><td></td><td>7 – Orange</td><td>590–640 nm</td><td></td><td></td></tr><tr><td>3 – Orange</td><td>590–640 nm</td><td></td><td></td><td>0 – CH₄-JST</td><td>614–624 nm</td><td></td><td></td></tr></table> - Principal investigator: Bradford Smith / University of Arizona (PDS/PRN website) - Data: PDS/PDI data catalog, PDS/PRN data catalog	Filters								Narrow-angle camera ^[30]				Wide-angle camera ^[31]				Name	Wavelength	Spectrum	Sensitivity	Name	Wavelength	Spectrum	Sensitivity	0 – Clear	280–640 nm			2 – Clear	280–640 nm			4 – Clear	280–640 nm			3 – Violet	350–450 nm			7 – UV	280–370 nm			1 – Blue	430–530 nm			1 – Violet	350–450 nm			6 – CH ₄ -U	536–546 nm			2 – Blue	430–530 nm			5 – Green	530–640 nm			5 – Green	530–640 nm			4 – Na-D	588–590 nm			6 – Green	530–640 nm			7 – Orange	590–640 nm			3 – Orange	590–640 nm			0 – CH ₄ -JST	614–624 nm
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Radio Science System (disabled)	(RSS)	Used the telecommunications system of the Voyager spacecraft to determine the physical properties of planets and satellites (ionospheres, atmospheres, masses, gravity fields, densities) and the amount and size distribution of material in Saturn's rings and the ring dimensions. - Principal investigator: G. Tyler / Stanford University PDS/PRN overview - Data: PDS/PPI data catalog, PDS/PRN data catalog (VG_2803), NSSDC data archive																																																																																						
<u>Infrared interferometer spectrometer and radiometer</u> (disabled)	(IRIS)	Investigates both global and local energy balance and atmospheric composition. Vertical temperature profiles are also obtained from the planets and satellites as well as the composition, thermal properties, and size of particles in Saturn's rings. Off to save power (June 3, 1998) - Principal investigator: Rudolf Hanel / NASA Goddard Space Flight Center (PDS/PRN website) - Data: PDS/PRN data catalog, PDS/PRN expanded data catalog (VGIRIS_0001, VGIRIS_002), NSSDC Jupiter data archive																																																																																						
<u>Ultraviolet Spectrometer</u> (disabled)	(UVS)	Designed to measure atmospheric properties, and to measure radiation. Off to save power (April 19, 2016) - Principal investigator: A. Broadfoot / University of Southern California (PDS/PRN website) - Data: PDS/PRN data catalog																																																																																						
<u>Triaxial Fluxgate Magnetometer</u> (active)	(MAG)	Designed to investigate the magnetic fields of Jupiter and Saturn, the interaction of the solar wind with the magnetospheres of these planets, and the magnetic field of interplanetary space out to the boundary between the solar wind and the magnetic field of interstellar space. One of two operational instruments. - Principal investigator: Norman F. Ness / NASA Goddard Space Flight Center (website) - Data: PDS/PPI data catalog, NSSDC data archive																																																																																						
<u>Plasma Spectrometer</u> (defective)	(PLS)	Investigates the microscopic properties of the plasma ions and measures electrons in the energy range from 5 eV to 1 keV. Off because of degraded performance (February 1, 2007) - Principal investigator: John Richardson / MIT (website) - Data: PDS/PPI data catalog, NSSDC data archive																																																																																						

Low Energy Charged Particle Instrument (disabled)	(LECP)	Measures the differential in energy fluxes and angular distributions of ions, electrons and the differential in energy ion composition. Off to save power (April 17, 2026), can still be reactivated - Principal investigator: Stamatios Krimigis / JHU / APL / University of Maryland (JHU/APL website / UMD website / KU website) - Data: UMD data plotting, PDS/PPI data catalog, NSSDC data archive
Cosmic Ray System (disabled)	(CRS)	Determines the origin and acceleration process, life history, and dynamic contribution of interstellar cosmic rays, the nucleosynthesis of elements in cosmic-ray sources, the behavior of cosmic rays in the interplanetary medium, and the trapped planetary energetic-particle environment. Off to save power (February 25, 2025) - Principal investigator: Edward Stone / Caltech / NASA Goddard Space Flight Center (website) - Data: PDS/PPI data catalog, NSSDC data archive
Planetary Radio Astronomy Investigation (disabled)	(PRA)	Uses a sweep-frequency radio receiver to study the radio-emission signals from Jupiter and Saturn. Off to save power (January 15, 2008) - Principal investigator: James Warwick / University of Colorado - Data: PDS/PPI data catalog, NSSDC data archive
Photopolarimeter System (defective)	(PPS)	Used a telescope with a polarizer to gather information on surface texture and composition of Jupiter and Saturn and information on atmospheric scattering properties and density for both planets. - Principal investigator: Arthur Lane / JPL (PDS/PRN website) - Data: PDS/PRN data catalog
Plasma Wave Subsystem (active)	(PWS)	Provides continuous, sheath-independent measurements of the electron-density profiles at Jupiter and Saturn as well as basic information on local wave–particle interaction, useful in studying the magnetosphere. One of two operational instruments. - Principal investigator: William Kurth / University of Iowa (website) - Data: PDS/PPI data catalog

Mission profile

Timeline of travel



Date	Event
1977-09-05	Spacecraft launched at 12:56:00 UTC.
1977-12-10	Entered <u>asteroid belt</u> .
1977-12-19	<i>Voyager 1</i> overtakes <i>Voyager 2</i> . (<i>see diagram</i>)
1978-09-08	Exited asteroid belt.
1979-01-06	Start Jupiter observation phase.
1979-03-05	Encounter with the Jovian system.
06:54	<i>Amalthea</i> flyby at 420,200 km.
12:05:26	<i>Jupiter</i> closest approach at 348,890 km from the center of mass.
15:14	<i>Io</i> flyby at 20,570 km.
18:19	<i>Europa</i> flyby at 733,760 km.
1979-03-06	
02:15	<i>Ganymede</i> flyby at 114,710 km.
17-08	<i>Callisto</i> flyby at 126,400 km.
1979-04-13	Phase end
1980-08-22	Start Saturn observation phase.
1980-11-12	Encounter with the Saturnian system.
05:41:21	<i>Titan</i> flyby at 6,490 km.
22:16:32	<i>Tethys</i> flyby at 415,670 km.
23:46:30	<i>Saturn</i> closest approach at 184,300 km from the center of mass.
1980-11-13	
01:43:12	<i>Mimas</i> flyby at 88,440 km.
01:51:16	<i>Enceladus</i> flyby at 202,040 km.
06:21:53	<i>Rhea</i> flyby at 73,980 km.
16:44:41	<i>Hyperion</i> flyby at 880,440 km.
1980-11-14	Phase end
1980-11-14	Begin extended mission.

Extended mission	
1990-02-14	Final images of the Voyager program acquired by <i>Voyager 1</i> to create the Solar System <i>Family Portrait</i> .
1998-02-17	<i>Voyager 1</i> overtakes <i>Pioneer 10</i> as the most distant spacecraft from the Sun, at 69.419 AU. <i>Voyager 1</i> is moving away from the Sun at over 1 AU per year faster than <i>Pioneer 10</i> .
2004-12-17	Passed the <u>termination shock</u> at 94 AU and entered the <u>heliosheath</u> .
2007-02-02	Terminated plasma subsystem operations.
2007-04-11	Terminated plasma subsystem heater.
2008-01-16	Terminated planetary radio astronomy experiment operations.
2012-08-25	Crossed the <u>heliopause</u> at 121 AU and entered interstellar space, becoming the first human-made object to exit the solar system. ^[32]
2014-07-07	Further confirmation probe is in <u>interstellar space</u> .
2016-04-19	Terminated Ultraviolet Spectrometer operations.
2017-11-28	"Trajectory correction maneuver" (TCM) thrusters are tested in their first use since November 1980. ^[33]
2023-11-14	Problems with onboard computer render it unable to send usable data back to Earth, engineers begin planning and developing a fix. ^{[34][35]}
2024-04-22	Engineers re-establish communication with the probe by moving code away from a broken memory chip in the FDS. ^[36]
2025-02-25	Turned off the cosmic ray subsystem. ^[37]
2026-04-17	Turned off Low-Energy Charged Particles instrument to save power (can still be reactivated). Only 2 instruments aboard operational as of now. ^[38]

Launch and trajectory



Voyager 1 lifted off atop a Titan IIIE.

The *Voyager 1* probe was launched on September 5, 1977, from Launch Complex 41 at the Cape Canaveral Air Force Station, aboard a Titan IIIE launch vehicle. The *Voyager 2* probe had been launched two weeks earlier, on August 20, 1977. Despite being launched later, *Voyager 1* reached both Jupiter^[39] and Saturn sooner, following a shorter trajectory.^[40]

Voyager 1's launch almost failed because Titan's LR-91 second stage shut down prematurely, leaving 1,200 pounds (540 kg) of propellant unburned. Recognizing the deficiency, the Centaur stage's on-board computers ordered a burn that was far longer than planned in order to compensate. Centaur extended its own burn and was able to give *Voyager 1* the additional velocity it

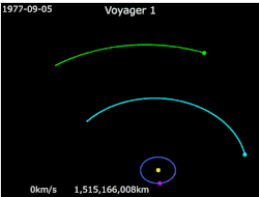
needed.^[41]

At cutoff, the Centaur was only 3.4 seconds from propellant exhaustion. If the same failure had occurred during *Voyager 2*'s launch a few weeks earlier, the Centaur would have run out of propellant before the probe reached the correct trajectory. Jupiter was in a more favorable position vis-à-vis Earth during the launch of *Voyager 1* than during the launch of *Voyager 2*.^[41]

Voyager 1's initial orbit had an aphelion of 8.9 AU (830 million mi), just a little short of Saturn's orbit of 9.5 AU (880 million mi). *Voyager 2*'s initial orbit had an aphelion of 6.2 AU (580 million mi), well short of Saturn's orbit.^[42]

Flyby of Jupiter

Voyager 1 began photographing Jupiter in January 1979. Its closest approach to Jupiter was on March 5, 1979, at a distance of about 349,000 kilometers (217,000 miles) from the planet's center.^[39] Because of the greater photographic resolution allowed by a closer approach, most observations of the moons, rings, magnetic fields, and the radiation belt environment of the Jovian



An animation of *Voyager 1*'s trajectory from September 1977 to December 31, 1981

■ *Voyager 1* · ■ Earth · ■ Jupiter · ■ Saturn · ■ Sun

system were made during the 48-hour period that bracketed the closest approach. Voyager 1 finished photographing the Jovian system in April 1979.^[43]

Information gathered by the *Pioneer 10* spacecraft helped engineers design *Voyager* to better cope with the intense radiation around Jupiter.^[44] Still, shortly before launch, strips of kitchen-grade aluminium foil were applied to certain cables to improve radiation shielding.^[45]

The discovery of ongoing volcanic activity on the moon Io was probably the greatest surprise. It was the first time active volcanoes had been seen on another body in the Solar System. It appears that activity on Io affects the entire Jovian system. Io appears to be the primary source of matter that pervades the Jovian magnetosphere – the region of space that surrounds the planet influenced by the planet's strong magnetic field. Sulfur, oxygen, and sodium, apparently erupted by Io's volcanoes and sputtered off the surface by the impact of high-energy particles, were detected at the outer edge of the magnetosphere of Jupiter.^[39] Once Voyager 1 passed within 5 *R*_J from the planet's center,^[46] it encountered the Io plasma torus.^[47] It received a radiation dosage one thousand times the lethal level for humans, the damage resulting in serious degradation of some high-resolution images of Io and Ganymede.^[48]

The two Voyager space probes made a number of important discoveries about Jupiter, its satellites, its radiation belts, and its never-before-seen planetary rings. Voyager 1 also discovered two new moons of Jupiter, the first one being Metis, was discovered orbiting just outside the ring, making it the first of Jupiter's moons to be identified by a spacecraft.^[49] The second new satellite, Thebe, was discovered between the orbits of Amalthea and Io.^[50]

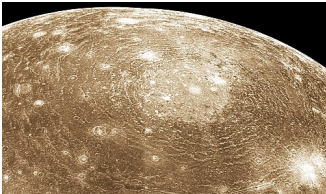
On February 25, 1979, when Voyager 1 was 9.2 million kilometers from Jupiter it transmitted the first detailed image of the Great Red Spot back to Earth. Cloud details as small as 160 km across were visible. The colorful, wavy cloud pattern seen to the west (left) of the GRS is the spot's wake region, where extraordinarily complex and variable cloud motions are observed.^[51]



Jupiter's Great Red Spot as seen from *Voyager 1*



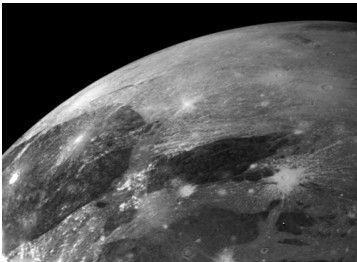
Voyager 1 composite image of Jupiter and its moons Europa (right) and Io (left) taken on the final day of *Voyager 1*'s Jupiter encounter.



Voyager 1 image image of Valhalla, a multi-ring impact structure 3,800 km in diameter on Callisto.



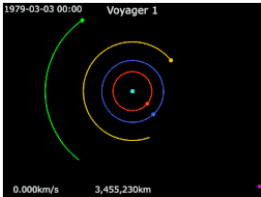
Voyager 1 image of Europa.



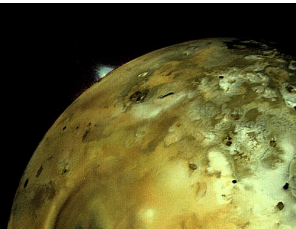
Voyager 1 image of Perrine Regio on Ganymede.



Voyager 1 mosaic covering Io's south polar region.



An animation of *Voyager 1*'s trajectory around Jupiter
Voyager 1 · Jupiter · Io · Europa · Ganymede · Callisto



Voyager 1 image of a volcanic eruption on Io 11 hours before closest approach.

[Media related to the Voyager 1 Jupiter encounter at Wikimedia Commons](#)

Flyby of Saturn

The gravitational assist trajectories at Jupiter were successfully carried out by both Voyagers, and the two spacecraft went on to visit Saturn and its system of moons and rings. *Voyager 1* encountered Saturn in November 1980, with the closest approach on November 12, 1980, when the space probe came within 124,000 kilometers (77,000 mi) of Saturn's cloud-tops. The space probe's cameras detected complex structures in the rings of Saturn, and its remote sensing instruments studied the atmospheres of Saturn and its giant moon Titan.^[52]

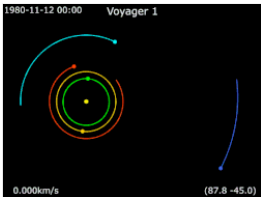
Voyager 1 found that about seven percent of the volume of Saturn's upper atmosphere is helium (compared with 11 percent of Jupiter's atmosphere), while almost all the rest is hydrogen. Since Saturn's internal helium abundance was expected to be the same as Jupiter's and the Sun's, the lower abundance of helium in the upper atmosphere may imply that the heavier helium may be slowly sinking through Saturn's hydrogen; that might explain the excess heat that Saturn radiates over energy it receives from the Sun. Winds blow at high speeds on Saturn. Near the equator, the Voyagers measured winds about 500 m/s (1,100 mph). The wind blows mostly in an easterly direction.^[40]

The Voyagers found aurora-like ultraviolet emissions of hydrogen at mid-latitudes in the atmosphere, and auroras at polar latitudes (above 65 degrees). The high-level auroral activity may lead to the formation of complex hydrocarbon molecules that are carried toward the equator. The mid-latitude auroras, which occur only in sunlit regions, remain a puzzle, since bombardment by electrons and ions, known to cause auroras on Earth, occurs primarily at high latitudes. Both Voyagers measured the rotation of Saturn (the length of a day) at 10 hours, 39 minutes, 24 seconds.^[52]

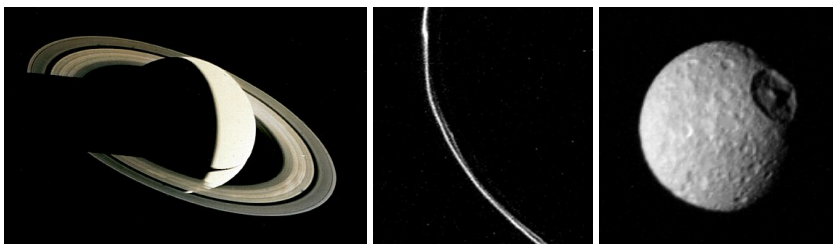
Voyager 1's mission included a flyby of Titan, Saturn's largest moon, which had long been known to have an atmosphere. Images taken by *Pioneer 11* in 1979 had indicated the atmosphere was substantial and complex, further increasing interest. The Titan flyby occurred as the spacecraft entered the system to avoid any possibility of damage closer to Saturn compromising observations, and approached to within 6,400 km (4,000 mi), passing behind Titan as seen from Earth and the Sun.^[53]

Voyager's measurement of the atmosphere's effect on sunlight and Earth-based measurement of its effect on the probe's radio signal were used to determine the atmosphere's composition, density, and pressure. Titan's mass was also measured by observing its effect on the probe's trajectory. The thick haze prevented any visual observation of the surface, but the measurement of the atmosphere's composition, temperature, and pressure led to speculation that lakes of liquid hydrocarbons could exist on the surface.^[53]

Because observations of Titan were considered vital, the trajectory chosen for *Voyager 1* was designed around the optimum Titan flyby, which took it below the south pole of Saturn and out of the plane of the ecliptic, ending its planetary science mission.^[54] Had *Voyager 1* failed or been unable to observe Titan, *Voyager 2*'s trajectory would have been altered to incorporate the Titan flyby,^{[53]:94} precluding any visit to Uranus and Neptune.^[7] The trajectory *Voyager 1* was launched into would not have allowed it to continue on to Uranus and Neptune,^{[54]:155} but could have been altered to avoid a Titan flyby and travel from Saturn to Pluto, arriving in 1986.^[9]



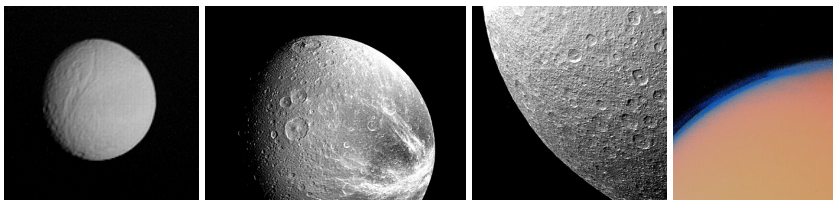
An animation of *Voyager 1* around Saturn
Voyager 1 · Saturn · Mimas · Enceladus · Tethys · Rhea · Titan



Crescent Saturn from 5.3 million km, four days after closest approach

Voyager 1 image of Saturn's narrow, twisted and braided F Ring.

Mimas at a range of 425,000 km. The crater Herschel is at upper right

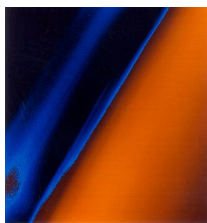


Tethys, with its giant rift valley Ithaca Chasma, from 1.2 million km.

Fractured 'wispy terrain' on Dione's trailing hemisphere.

The icy surface of Rhea is nearly saturated with impact craters.

Titan's thick haze layer is shown in this enhanced Voyager 1 image.

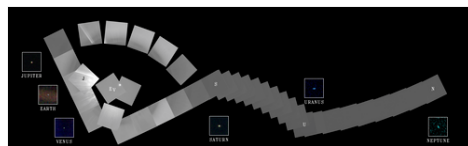


Layers of haze, composed of complex organic compounds, covering Saturn's satellite Titan.

 Media related to the Voyager 1 Saturn encounter at Wikimedia Commons

Exit from the heliosphere

On February 14, 1990, *Voyager 1* took the first "family portrait" of the Solar System as seen from outside,^[56] which includes the image of planet Earth known as *Pale Blue Dot*. Soon afterward, its cameras were deactivated to conserve energy and computer resources for other equipment. The camera software has been removed from the spacecraft, so it would now be complex to get them working again. Earth-side software and computers for reading the images are also no longer available.^[7]



The *Family Portrait* of the Solar System acquired by *Voyager 1*, February 14, 1990

On February 17, 1998, *Voyager 1* reached a distance of 69 AU (6.4 billion mi; 10.3 billion km) from the Sun and overtook *Pioneer 10* as the most distant spacecraft from Earth.^{[57][58]} Traveling at about 17 km/s (11 mi/s), it has the fastest heliocentric recession speed of any spacecraft.^[59]

As *Voyager 1* headed for interstellar space, its instruments continued to study the Solar System. Jet Propulsion Laboratory scientists used the plasma wave experiments aboard *Voyager 1* and 2 to look for the heliopause, the boundary at which the solar wind transitions into the interstellar medium.^[60] As of 2013, the probe was moving with a relative velocity to the Sun of about 61,197 kilometers per hour (38,026 mph).^[43] With the velocity the probe is currently maintaining, *Voyager 1* is traveling about 523 million km (325 million mi) per year,^[61] or about one light-year per 18,000 years.

Termination shock

Scientists at the Johns Hopkins University Applied Physics Laboratory believe that *Voyager 1* entered the termination shock in February 2003.^[62] This marks the point where the solar wind slows to subsonic speeds. Some other scientists expressed doubt and discussed this in the journal *Nature* of November 6, 2003.^[63] The issue would not be resolved until other data became available, since *Voyager 1*'s solar-wind detector ceased functioning in 1990. This failure meant that termination shock detection would have to be inferred from the data from the other instruments on board.^{[64][65][66]}

In May 2005, a NASA press release said that the consensus was that *Voyager 1* was then in the heliosheath.^[67] In a scientific session at the American Geophysical Union meeting in New Orleans, Louisiana, on May 25, 2005, Ed Stone presented evidence that the craft crossed the termination shock in late 2004.^[68] This event is estimated to have occurred on December 15, 2004, at a distance of 94 AU (8,700 million mi) from the Sun.^{[68][69]}

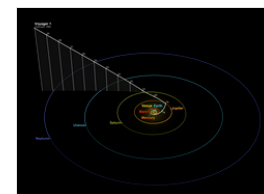
Heliosheath

On March 31, 2006, amateur radio operators from AMSAT in Germany tracked and received radio waves from *Voyager 1* using the 20-meter (66 ft) dish at Bochum with a long integration technique. Retrieved data was checked and verified against data from the Deep Space Network station at Madrid, Spain. This seems to be the first such amateur tracking of *Voyager 1*.^[70]

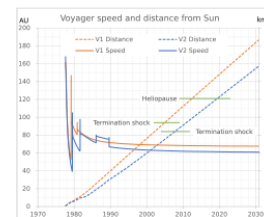
It was confirmed on December 13, 2010, that *Voyager 1* had passed the reach of the radial outward flow of the solar wind, as measured by the Low Energy Charged Particle device. It is suspected that solar wind at this distance turns sideways because of interstellar wind pushing against the heliosphere. Since June 2010, detection of solar wind had been consistently at zero, providing conclusive evidence of the event.^{[71][72]} On this date, the spacecraft was approximately 116 AU (17.4 billion km; 10.8 billion mi) from the Sun.^[73]

Voyager 1 was commanded to change its orientation to measure the sideways motion of the solar wind at that location in space in March 2011 (~33 years 6 months from launch). A test roll in February had confirmed the spacecraft's ability to maneuver and reorient itself. The course of the spacecraft was not changed. It rotated 70 degrees counterclockwise with respect to Earth to detect the solar wind. This was the first time the spacecraft had done any major maneuvering since the *Family Portrait* photograph of the planets was taken in 1990. After the first roll the spacecraft had no problem in reorienting itself with Alpha Centauri, *Voyager 1*'s guide star, and it resumed sending transmissions back to Earth. *Voyager 1* was expected to enter interstellar space "at any time". *Voyager 2* was still detecting outward flow of solar wind at that point but it was estimated that in the following months or years it would experience the same conditions as *Voyager 1*.^{[74][75]}

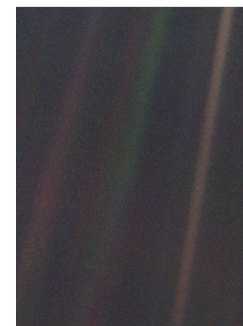
The spacecraft was reported at 12.44° declination and 17.163 hours right ascension, and at an ecliptic latitude of 34.9° (the ecliptic latitude changes very slowly), placing it in the constellation Ophiuchus as observed from the Earth on May 21, 2011.^[7]



The position of *Voyager 1* above the plane of the ecliptic on February 14, 1990, the day *Family Portrait* was taken.



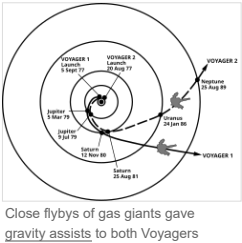
Voyager 1 and 2 speed and distance from the Sun



The *Pale Blue Dot* image showing Earth from 6 billion kilometers (3.7 billion miles) appearing as a tiny dot (the bluish-white speck approximately halfway down the light band to the right) within the darkness of deep space.^[55]

On December 1, 2011, it was announced that *Voyager 1* had detected the first Lyman-alpha radiation originating from the Milky Way galaxy. Lyman-alpha radiation had previously been detected from other galaxies, but because of interference from the Sun, the radiation from the Milky Way was not detectable.^[76]

NASA announced on December 5, 2011, that *Voyager 1* had entered a new region referred to as a "cosmic purgatory". Within this stagnation region, charged particles streaming from the Sun slow and turn inward, and the Solar System's magnetic field is doubled in strength as interstellar space appears to be applying pressure. Energetic particles originating in the Solar System decline by nearly half, while the detection of high-energy electrons from outside increases 100-fold. The inner edge of the stagnation region is located approximately 113 AU from the Sun.^[77]



Heliopause

NASA announced in June 2012 that the probe was detecting changes in the environment that were suspected to correlate with arrival at the heliopause.^[78] *Voyager 1* had reported a marked increase in its detection of charged particles from interstellar space, which are normally deflected by the solar winds within the heliosphere from the Sun. The craft thus began to enter the interstellar medium at the edge of the Solar System.^[79]

Voyager 1 became the first spacecraft to cross the heliopause in August 2012, then at a distance of 121 AU (1.12×10^{10} mi; 1.81×10^{10} km) from the Sun, although this was not confirmed for another year.^{[80][81][82][83][84]}

As of September 2012, sunlight took 16.89 hours to reach *Voyager 1*, which was at a distance of 121 AU. The apparent magnitude of the Sun from the spacecraft was −16.3 (about 30 times brighter than the full Moon).^[85] The spacecraft was traveling at 17.043 km/s (10,590 mi/s) relative to the Sun. At this rate, it would need about 17,565 years to travel a single light-year.^[85] To compare, Proxima Centauri, the closest star to the Sun, is about 4.2 light-years (2.65×10^5 AU) distant. If the spacecraft was traveling in the direction of that star, it would take 73,775 years to reach it. *Voyager 1* is heading in the direction of the constellation Ophiuchus.^[85]

In late 2012, researchers reported that particle data from the spacecraft suggested that the probe had passed through the heliopause. Measurements from the spacecraft revealed a steady rise since May in collisions with high energy particles (above 70 MeV), which are thought to be cosmic rays emanating from supernova explosions far beyond the Solar System, with a sharp increase in these collisions in late August. At the same time, in late August, there was a dramatic drop in collisions with low-energy particles, which are thought to originate from the Sun.^[86]

Ed Roelof, a space scientist at Johns Hopkins University and principal investigator for the Low-Energy Charged Particle instrument on the spacecraft, declared that "most scientists involved with *Voyager 1* would agree that [these two criteria] have been sufficiently satisfied".^[86] However, the last criterion for officially declaring that *Voyager 1* had crossed the boundary, the expected change in magnetic field direction (from that of the Sun to that of the interstellar field beyond), had not been observed (the field had changed direction by only 2 degrees),^[81] which suggested to some that the nature of the edge of the heliosphere had been misjudged.

On December 3, 2012, Voyager project scientist Ed Stone of the California Institute of Technology said, "Voyager has discovered a new region of the heliosphere that we had not realized was there. We're still inside, apparently. But the magnetic field now is connected to the outside. So it's like a highway letting particles in and out."^[87] The magnetic field in this region was 10 times more intense than *Voyager 1* encountered before the termination shock. It was expected to be the last barrier before the spacecraft exited the Solar System completely and entered interstellar space.^{[88][89][90]}

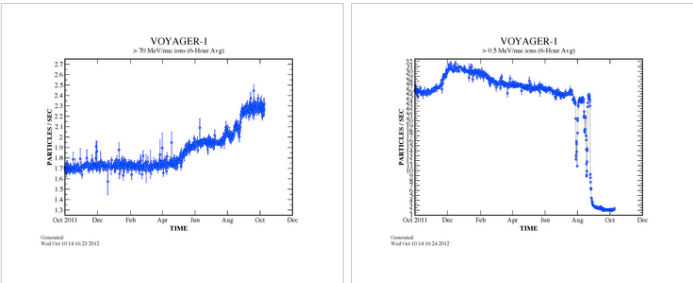
Interstellar medium

In March 2013, it was announced that *Voyager 1* might have become the first spacecraft to enter interstellar space, having detected a marked change in the plasma environment on August 25, 2012. However, until September 12, 2013, it was still an open question as to whether the new region was interstellar space or an unknown region of the Solar System. At that time, the former alternative was officially confirmed.^{[91][92]}

In 2013, *Voyager 1* was exiting the Solar System at a speed of about 3.6 AU (330 million mi; 540 million km) per year, which is 61,602 km/h, 4.83 times the diameter of Earth (12,742 km) per hour; whereas *Voyager 2* is going slower, leaving the Solar System at 3.3 AU (310 million mi; 490 million km) per year.^[93] Each year, *Voyager 1* increases its lead over *Voyager 2*.

Voyager 1 reached a distance of 135 AU (12.5 billion mi; 20.2 billion km) from the Sun on May 18, 2016.^[4] On September 5, 2017, that had increased to about 139.64 AU (12.980 billion mi; 20.890 billion km) from the Sun, or just over 19 light-hours; at that time, *Voyager 2* was 115.32 AU (10.720 billion mi; 17.252 billion km) from the Sun.^[4]

Its progress can be monitored at NASA's website.^{[4][94]}



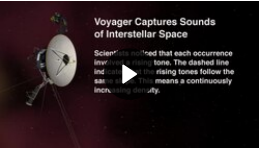
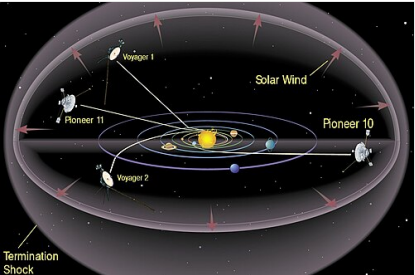
On September 12, 2013, NASA confirmed that *Voyager 1* had reached the interstellar medium in August 2012 as previously observed. The generally accepted date of arrival is August 25, 2012 (approximately 10 days before the 35th anniversary of its launch), the date durable changes in the density of energetic particles were first detected.^{[82][83][84]} By this point, most space scientists had abandoned the hypothesis that a change in magnetic field direction must accompany a crossing of the heliopause;^[83] a new model of the heliopause predicted that no such change would be found.^[95]

A key finding that persuaded many scientists that the heliopause had been crossed was an indirect measurement of an 80-fold increase in electron density, based on the frequency of plasma oscillations observed beginning on April 9, 2013,^[83] triggered by a solar outburst that had occurred in March 2012.^[80] Electron density is expected to be two orders of magnitude higher outside the heliopause than within.^[82]

Weaker sets of oscillations measured in October and November 2012^{[92][96]} provided additional data. An indirect measurement was required because *Voyager 1*'s plasma spectrometer had stopped working in 1980.^[84] In September 2013, NASA released recordings of audio transductions of these plasma waves, the first to be measured in interstellar space.^[97]

While *Voyager 1* is commonly spoken of as having left the Solar System simultaneously with having left the heliosphere, the two are not the same. The Solar System is usually defined as the vastly larger region of space populated by bodies that orbit the Sun. The craft is presently less than one-seventh the distance to the aphelion of Sedna, and it has not yet entered the postulated position of the Oort cloud, the hypothetical source region of long-period comets, regarded by astronomers as the outermost zone of the Solar System.^{[81][92]}

In October 2020, astronomers reported a significant unexpected increase in density in the space beyond the Solar System as detected by the *Voyager 1* and *Voyager 2* space probes. According to the researchers, this implies that "the density gradient is a large-scale feature of the VLISM (very local interstellar medium) in the general direction of the heliospheric nose".^{[98][99]}



Voyager 1 transmitted audio signals generated by plasma waves from interstellar space

In May 2021, NASA reported on the continuous measurement, for the first time, of the density of material in interstellar space and, as well, the detection of interstellar sounds for the first time.^[100]

Future of the probe

Remaining lifespan

In December 2017, NASA fired all four of *Voyager 1*'s trajectory correction maneuver (TCM) thrusters for the first time since 1980. The TCM thrusters were used in the place of a degraded set of jets to help keep the probe's antenna pointed towards Earth. Using the TCM thrusters allowed *Voyager 1* to continue transmitting data to NASA for two to three more years.^{[103][33]}

Due to the diminishing electrical power available, the Voyager team has had to prioritize which instruments to keep on and which to turn off. Heaters and other spacecraft systems have been turned off one by one as part of power management. The fields and particles instruments that are the most likely to send back key data about the heliosphere and interstellar space have been prioritized to keep operating. In 2023, engineers expected the spacecraft to continue operating at least one science instrument until around 2025.^[104] As of late April 2026, two instruments remain in operation.^[38]

Year	End of specific capabilities as a result of the available electrical power limitations
1998	Termination of Ultraviolet Spectrometer (UVS) ^[4]
2007	Termination of plasma subsystem (PLS) ^[105]
2008	Power off Planetary Radio Astronomy Experiment (PRA) ^[105]
2016	Termination of scan platform and Ultraviolet Spectrometer (UVS) observations ^[4]
2025	Termination of Cosmic Ray Subsystem (CRS) ^[37]
2026	Termination of Low Energy Charged Particle Instrument. ^[37]
Unknown date	Termination of Data Tape Recorder (DTR) operations (limited by ability to capture 1.4 kbit/s data using a 70 m/34 m antenna array; this is the minimum rate at which the DTR can read out data). ^[105]
Unknown date	Termination of gyroscopic operations (previously 2017, but backup thrusters active for continuation of gyroscopic operations). ^[105]
2027–2036	Will no longer be able to power even a single instrument. After 2036, both probes will be out of range of the Deep Space Network . ^[16]

Concerns with the orientation thrusters

Some thrusters needed to control the attitude of the spacecraft and point its high-gain antenna in the direction of Earth are out of use due to clogging problems in their hydrazine lines. The spacecraft no longer has a backup available for its thruster system and "everything onboard is single-string," according to Suzanne Dodd, Voyager project manager at JPL, in an interview with *Ars Technica*.^[106] NASA has accordingly decided to modify the spacecraft's computer software in order to reduce the rate at which the hydrazine lines clog. NASA will first deploy the modified software on *Voyager 2*, which is less distant from Earth, before deploying it on *Voyager 1*.^[106]

In September 2024, NASA performed a "thruster swap", switching from a clogged set of thrusters to less clogged ones that had not been used since 2018.^[107]

In May 2025, the team was able to revive the backup thrusters used for roll motion that were unusable since 2004.^[14]

Interstellar velocity (<i>v</i> _∞)	
Probe	Velocity (<i>v</i> _∞)
<i>Pioneer 10</i>	11.8 km/s (2.49 au/yr)
<i>Pioneer 11</i>	11.1 km/s (2.34 au/yr)
<i>Voyager 1</i>	16.9 km/s (3.57 au/yr) ^[101]
<i>Voyager 2</i>	15.2 km/s (3.21 au/yr)
<i>New Horizons</i>	12.6 km/s (2.66 au/yr)



An image of *Voyager 1*'s radio signal on February 21, 2013.^[102]

Communication issues

In May 2022, NASA reported that *Voyager 1* had begun transmitting "mysterious" and "peculiar" telemetric data to the Deep Space Network (DSN). It confirmed that the operational status of the craft remained unchanged, but that the issue stemmed from the Attitude Articulation and Control System (AACS). NASA's Jet Propulsion Laboratory published a statement on May 18, 2022, that the AACS was functional but sending invalid data.^{[108][109]}

The problem was eventually traced to the AACS sending its telemetry through a computer that had been non-operational for years, resulting in data corruption. In August 2022, NASA transmitted a command to the AACS to use another computer, which resolved the problem. An investigation into what caused the initial switch is underway, though engineers have hypothesized that the AACS had executed a bad command from another onboard computer.^{[110][111]}

Voyager 1 began transmitting unreadable data on November 14, 2023. On December 12, 2023, NASA announced that *Voyager 1*'s flight data system was unable to use its telemetry modulation unit, preventing it from transmitting scientific data.^[112] On March 24, 2024, NASA announced that they had made significant progress on interpreting the data being received from the spacecraft.^[113] Engineers reported in April 2024 that the failure was likely in a memory bank of the Flight Data Subsystem (FDS), one of the three onboard computer systems, probably from being struck by a high-energy particle or that it simply wore out due to age. The FDS was not communicating properly with the telemetry modulation unit (TMU), which began transmitting a repeating sequence of ones and zeros indicating that the system was in a stuck condition. After a reboot of the FDS, communications remained unusable.^[114]

The probe still received commands from Earth, and was sending a carrier tone indicating it was operational. Commands sent to alter the modulation of the tone succeeded, confirming that the probe was still responsive.^[115] The *Voyager* team began developing a workaround,^{[116][117]} and on April 20 communication of health and status was restored by rearranging code away from the defective FDS memory chip, three percent of which was corrupted beyond repair.^{[36][118]}

Because the memory is corrupt, the code needed to be relocated, but there was no place for an extra 256 bits; the spacecraft's total memory is only 68 kilobytes. To make it work, the engineers deleted unused code, for example the code used to transmit the data from Jupiter, that cannot be used at the current transmission rate. All the data from the "anomaly period" is lost.^[119] On May 22, NASA announced that Voyager 1 "resumed returning science data from two of its four instruments", with work towards the others ongoing.^[120] On June 13, NASA confirmed that the probe returns data from all four instruments.^[121]

In October 2024, the probe turned off its X-band radio transmitter that was used for communications with the DSN. It was caused by the probe's fault protection system that was activated after NASA turned on one of the heaters on October 16. The fault protection system lowered the transmission rate, but the engineers were able to find the signal. Later, on October 19, the transmission stopped; the fault protection system was triggered once again and switched to the S-band transmitter, that was previously used in 1981.^[122] NASA reported that the team reactivated the X-band transmitter and then resumed collecting data in mid-November.^[123]

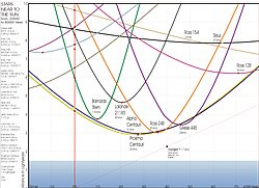
Between May 2025 and February 2026, Deep Space Station 43 in Canberra, Australia—the only antenna capable of sending commands to Voyager 1 and 2—was offline for major upgrades, with only limited operational windows in August and December 2025. This created a critical deadline for engineers to revive backup roll thrusters before full loss of communication.^[14]

Far future

Provided *Voyager 1* does not collide with anything and is not retrieved, it is expected to reach the theorized Oort cloud in about 300 years^{[124][125]} and take about 30,000 years to pass through it.^{[81][92]}

Though it is not heading towards any particular star, in about 40,000 years, it will pass within 1.6 light-years (0.49 parsecs; 100,000 astronomical units) of the star Gliese 445, which is in the constellation Camelopardalis and 17.1 light-years from Earth (as of 2010).^[126] That star is generally moving towards the Solar System at about 119 km/s (430,000 km/h; 270,000 mph).^[126]

In 300,000 years, it will pass within less than 1 light-year of the M3V star TYC 3135–52–1.^[127]



In about 50,000 years *Voyager 1* will be as distant as several nearby stars

The *New Horizons* space probe will never pass it, despite being launched from Earth at a higher speed than either Voyager spacecraft. The Voyager spacecraft benefited from multiple planetary flybys to increase their heliocentric velocities, whereas *New Horizons* received only a single such boost, from its Jupiter flyby in 2007. As of 2018, *New Horizons* is traveling at about 14 km/s (8.7 mi/s), 3 km/s (1.9 mi/s) slower than *Voyager 1*, and *New Horizons*, being closer to the Sun, is slowing more rapidly.^[128] NASA says that: "The Voyagers are destined – perhaps eternally – to wander the Milky Way."^[129]

Golden record

Both Voyager space probes carry a gold-plated audio-visual disc, a compilation meant to showcase the diversity of life and culture on Earth in the event that either spacecraft is ever found by any extraterrestrial discoverer.^{[130][131]} The record, made under the direction of a team including Carl Sagan and Timothy Ferris, includes photos of the Earth and its lifeforms, a range of scientific information, spoken greetings from people such as the Secretary-General of the United Nations (Kurt Waldheim) and a medley, "Sounds of Earth".^[132]

It includes the sounds of whales, a baby crying, waves breaking on a shore, and a collection of music spanning different cultures and eras including works by Wolfgang Amadeus Mozart, Blind Willie Johnson, Chuck Berry and Valya Balkanska. Other Eastern and Western classics are included, as well as performances of indigenous and folk music from around the world. The record also contains greetings in 55 different languages.^[132] The project aimed to portray the richness of life on Earth and stand as a testament to human creativity and the desire to connect with the cosmos.^{[131][132]}



Voyager Golden Record



A child's greeting (the voice of Nick Sagan) in English recorded on the Voyager Golden Record

See also

- The Farthest*, a 2017 documentary on the Voyager program
- Interstellar probe
- List of artificial objects leaving the Solar System
- List of missions to the outer planets
- Local Interstellar Cloud
- Space exploration
- Specific orbital energy of *Voyager 1*
- Timeline of artificial satellites and space probes

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